

2 Supervised learning

$$L(\phi) := \sum_{i=1}^1 (f(x_i, \phi) - y_i)^2 = \sum_{i=1}^1 (\phi_0 + \phi_1 x_i - y_i)^2$$

$$\begin{aligned} 2.1 \quad \frac{\partial L}{\partial \phi_0} &= \frac{\partial \sum_{i=1}^1 (\phi_0 + \phi_1 x_i - y_i)^2}{\partial \phi_0} = \sum_{i=1}^1 (\phi_0 + \phi_1 x_i - y_i) 2 \cdot \underbrace{\frac{\partial (\phi_0 + \phi_1 x_i - y_i)^2}{\partial \phi_0}}_{=2} \\ &= 2 \sum_{i=1}^1 (\phi_0 + \phi_1 x_i - y_i) = 2 | \phi_0 + 2 \sum_{i=1}^1 \phi_1 x_i - 2 \sum_{i=1}^1 y_i \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial \phi_1} &= \sum_{i=1}^1 \left((\phi_0 + \phi_1 x_i - y_i) \cdot 2 \cdot \underbrace{\frac{\partial (\phi_0 + \phi_1 x_i - y_i)}{\partial \phi_1}}_{=x_i} \right) \\ &= 2 \phi_0 \sum_{i=1}^1 x_i + 2 \phi_1 \sum_{i=1}^1 x_i^2 - 2 \sum_{i=1}^1 y_i x_i \end{aligned}$$

$$2.2 \quad \frac{\partial L}{\partial \phi_0} \stackrel{!}{=} 0 \Leftrightarrow \phi_0 = \frac{-2 \sum_{i=1}^1 \phi_1 x_i + 2 \sum_{i=1}^1 y_i}{2 |} = \frac{\sum_{i=1}^1 \phi_1 x_i + \sum_{i=1}^1 y_i}{1}$$

$$\frac{\partial L}{\partial \phi_1} \stackrel{!}{=} 0 \Leftrightarrow \phi_1 = \frac{-\cancel{2} \phi_0 \sum_{i=1}^1 x_i + \cancel{2} \sum_{i=1}^1 y_i x_i}{\cancel{2} \sum_{i=1}^1 x_i}$$

We solve for the parameter we took the partial derivative of.

$$\begin{aligned} 2.3 \quad x = g(y, \phi) &:= \phi_0 + \phi_1 y \Rightarrow L(\phi) = \sum_{i=1}^1 (g(y_i, \phi) - x_i)^2 \\ &= \sum_{i=1}^1 (\phi_0 + \phi_1 y_i - x_i)^2 \end{aligned}$$

$$y = g^{-1}(x, \phi) := \frac{-\phi_0 + x}{\phi_1} \Rightarrow \begin{array}{l} x \text{ from } y: \text{ horizontal mean is minimized!} \\ y \text{ from } x: \text{ vertical mean is minimized!} \end{array}$$

! It would not make the same predictions if !

! Here is no perfect line through the data,
as these are different optimization problems